# Lingzhi Yuan

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## EDUCATION

**University of Maryland, College Park** Aug. 2025 – Present
*Ph.D. in Computer Science (expected May 2030)*
Advisor: Prof. Furong Huang

**Zhejiang University** Sep. 2021 – Jun. 2025
*B.Eng. in Automation* | *GPA: 3.96/4.00* | *Rank: 1/58*
Minor in Intensive Training Honors Program (ITP), Chu Kochen Honors College (Top 40/5400)

## EXPERIENCE

**Furong Huang Lab, University of Maryland** Aug. 2025 – Present
*Research Assistant* | Trustworthy ML and LLM/agent safety, advised by Prof. Furong Huang.

**UNIUBI AI, Hangzhou, China** Mar. 2025 – Jun. 2025
*Student Intern* | Instruction following and data augmentation for vision-language models.

**Secure Learning Lab, The University of Chicago** Mar. 2024 – Dec. 2024
*Research Assistant* | Trustworthy ML and multimodal safety, advised by Prof. Bo Li.

**Nanyang Technological University, Singapore** Aug. 2023 – Sep. 2023
*Summer School Participant* | Machine Learning and Its Applications program, led by Prof. Kezhi Mao.

## SELECTED PUBLICATIONS

* denotes equal contribution

**Meta$^2$: Precomputing Reusable Knowledge for Online Harness Evolution in Long-Horizon Agents**
Lingzhi Yuan*, Yiguang Cai*, Weize Liu, Feijiang Han, Yongkyun Lee, Udari Madhushani Sehwag, Furong Huang
*arXiv preprint*

**FlowBank: Query-Adaptive Agentic Workflow Optimization through Precompute-and-Reuse**
Lingzhi Yuan, Chenghao Deng, Fangxu Yu, Souradip Chakraborty, Mohammad Rostami, Furong Huang
*Conference on Neural Information Processing Systems (NeurIPS), 2026*

**PromptGuard: Soft Prompt-Guided Unsafe Content Moderation for Text-to-Image Models**
Lingzhi Yuan*, Xinfeng Li*, Chejian Xu, Guanhong Tao, Xiaojun Jia, Yihao Huang, Wei Dong, Yang Liu, Bo Li
*IEEE Transactions on Information Forensics and Security (TIFS)*

**TSRBench: A Comprehensive Multi-task Multi-modal Time Series Reasoning Benchmark for Generalist Models**
Fangxu Yu, Xingang Guo, Lingzhi Yuan, Haoqiang Kang, Hongyu Zhao, Lianhui Qin, Furong Huang, Bin Hu, Tianyi Zhou
*International Conference on Machine Learning (ICML), 2026*

**MMDT: Decoding the Trustworthiness and Safety of Multimodal Foundation Models**
Chejian Xu*, Jiawei Zhang*, Zhaorun Chen*, Chulin Xie*, Mintong Kang*, Zhuowen Yuan*, Zidi Xiong*, Chenhui Zhang, Lingzhi Yuan, Yi Zeng, Peiyang Xu, Chengquan Guo, Andy Zhou, Jeffrey Ziwei Tan, Zhun Wang, Alexander Xiong, Xuandong Zhao, Yu Gai, Francesco Pinto, Yujin Potter, Zhen Xiang, Zinan Lin, Dan Hendrycks, Dawn Song, Bo Li
*International Conference on Learning Representations (ICLR), 2025*

**Hypothesis-Conditioned Evaluation for Efficient Harness Optimization**
Lingzhi Yuan*, Zhi Zhang*, Arash A. Amini, Furong Huang
*Under review*

**LatentShield: Composable Latent Multi-Agent Systems with Decision-Boundary Monitoring**
Baicheng Chen*, Anirudh Satheesh*, Lingzhi Yuan, Drew Gjerstad, Andrew Mendz, Ami Britti Antoni Gnanaraj, Mingyu Kang, Benjamin Leinwand, Andrew Adams Schoen, Furong Huang
*Under review*

## SELECTED PROJECTS

**Query-Adaptive Agentic Workflow Optimization** Sep. 2025 – May 2026

*Project Lead* | *Advisor: Prof. Furong Huang, University of Maryland*

A portfolio-based framework that precomputes a bank of complementary agentic workflows offline and routes each query to the best one at inference time.

- Showed workflow complementarity: discarded task-level workflows retain reuse value, and query-level generation gains can be recovered by cheaper precomputed workflows.
- Designed a three-stage pipeline: complementarity-oriented MCTS search, coverage-aware portfolio curation, and GNN-based query–workflow matching.
- Best average score across five benchmarks, improving over the strongest automated and handcrafted baselines by 4.26% and 14.92% (relative).

**Soft Prompt-Guided Defense for Text-to-Image Models** Jun. 2024 – Dec. 2024

*Project Lead* | *Advisor: Prof. Bo Li, University of Chicago / UIUC*

An effective and efficient framework to mitigate NSFW generation across unsafe categories in T2I models.

- Surveyed safety protection techniques for T2I models and identified key open challenges.
- Built a multi-category moderation framework achieving state-of-the-art NSFW moderation with high time efficiency; published in IEEE TIFS.

**Comprehensive Assessment of Trustworthiness in Multimodal Models** Mar. 2024 – Oct. 2024

*Research Assistant* | *Advisor: Prof. Bo Li, University of Chicago / UIUC*

Evaluated the safety and trustworthiness of multimodal foundation models.

- Applied red-teaming algorithms (MMP, GCG for T2I; Attack-VLM for I2T) on surrogate models to build an adversarial robustness benchmark.
- Evaluated state-of-the-art models (e.g., DALL-E 3, GPT-4o) and analyzed their vulnerability under adversarial conditions; published at ICLR 2025.

## SKILLS

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|---|---|
| **Programming** | Python, C++, MATLAB |
| **Tools** | PyTorch, Transformers, Diffusers, OpenCV, Git, Linux/Unix |
| **Languages** | TOEFL 105 (Reading 30, Listening 27, Speaking 23, Writing 25) |

## HONORS AND AWARDS

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|---|---|
| University of Maryland Graduate School Dean's Fellowship | 2025 |
| Infineon Power Semiconductor Scholarship | 2023, 2024 |
| Zhejiang University Second Class Scholarship | 2022, 2023 |
| Nitori International Scholarship | 2022 |

## SERVICES

**Teaching Assistant**, CMSC 320: Introduction to Data Science, University of Maryland Fall 2025

**Journal Reviewer**, International Journal of Computer Vision (IJCV)

**Conference Reviewer**, International Conference on Learning Representations (ICLR) 2027